

COMPLEX ANALYSIS

Complex Numbers:

The set of complex number is $C = \{a + ib : a, b \in \mathbb{R}, i = \sqrt{-1}\}$.

Note: If $Z = a + ib$ then

$\text{Re}(Z) = \text{real part of } Z = a$

$\text{Im}(Z) = \text{imaginary part of } Z = b$

Properties of i:

Following are the some useful propertis of i.

- $(i)^1 = i, (i)^2 = -1, (i)^3 = -i, (i)^4 = 1$
- Sum of consecutive four power of i is zero.
for example $i^{12} + i^{13} + i^{14} + i^{15} = 0$.
in general $i^{1+n} + i^{2+n} + i^{3+n} + i^{4+n} = 0$ for some fixed $n \in \mathbb{Z}$
- Product of Consecutive Four Power of i is -1 .
for example $i^{11} \times i^{12} \times i^{13} \times i^{14} = -1$
in general $i^{1 \times n} + i^{2 \times n} + i^{3 \times n} + i^{4 \times n} = 0$ for some fixed $n \in \mathbb{Z}$.
- Sum of consecutive four multiple power of i is zero.
for example $i^5 \times i^{10} \times i^{15} \times i^{20} = -1$
in general $i^n + i^{2n} + i^{3n} + i^{4n} = 0$ for some fixed $n \in \mathbb{Z}$.
- Product of contraction four multiple power of i is -1 .
for example $i^3 \times i^6 \times i^9 \times i^{12} = -1$
in general $i^n \times i^{2n} \times i^{3n} \times i^{4n} = 0$ for some fixed $n \in \mathbb{Z}$.

Algebra of Complex Numbers:

Following table shows the information of algebra of complex numbers.

Complex Numbers	$Z_1 = a + ib, \quad Z_2 = c + id$
Addition	$Z_1 + Z_2 = (a + c) + i(b + d)$
Subtraction	$Z_1 - Z_2 = (a - c) + i(b - d)$
Multiplication	$Z_1 Z_2 = (ac - bd) + i(bc + ad)$
Division	$\frac{Z_1}{Z_2} = \frac{(ac + bd) + i(bc - ad)}{c^2 + d^2}$

Conjugate of Complex Number:

Conjugate of a complex number $z = a + ib$ is defined as $\bar{z} = a - ib$.

For example, $\overline{4 + 5i} = 4 - 5i$ and $\overline{4 - 5i} = 4 + 5i$.

Properties of Conjugate:

- (i) $\overline{(\bar{z})} = z$
- (ii) $z = \bar{z}$ if and only if z is purely real
- (iii) $z = -\bar{z}$ if and only if z is purely imaginary
- (iv) $z + \bar{z} = 2 \operatorname{Re}(z)$ and $z - \bar{z} = 2i \operatorname{Im}(z)$
- (v) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
- (vi) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$
- (vii) $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$
- (viii) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, z_2 \neq 0$
- (ix) if $z = f(z_1)$, then $\bar{z} = f(\bar{z}_1)$
- (x) $\overline{(z^n)} = (\bar{z})^n$
- (xi) $z_1 \bar{z}_2 + \bar{z}_1 z_2 = 2 \operatorname{Re}(\bar{z}_1 z_2) = 2 \operatorname{Re}(z_1 \bar{z}_2)$

Modulus of a Complex Number:

Modulus of a complex number $z = a + ib$, denoted as

$$|z| = \sqrt{a^2 + b^2}, \text{ where } a = \operatorname{Re}(z), b = \operatorname{Im}(z),$$

Sometimes, $|z|$ is called absolute value of z . Note that $|z| \geq 0$.

For example, if $z = 3 + 2i$, then $|z| = \sqrt{3^2 + 2^2} = \sqrt{13}$

Properties of Modulus:

- (i) $|z| \geq 0$ and $|z| = 0$ if and only if $z = 0$, i. e., $x = 0, y = 0$
- (ii) $|z| = |\bar{z}| = |-\bar{z}| = |-\bar{z}|$
- (iii) $z\bar{z} = |z|^2$
- (iv) $-|z| \leq \operatorname{Re}(z) \leq |z|$ and $-|z| \leq \operatorname{Im}(z) \leq |z|$
- (v) $|z^n| = |z|^n$
- (vi) $|z_1 z_2| = |z_1| |z_2|$

$$(vii) \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$(viii) \quad |z_1 + \pm z_2| \leq |z_1| + |z_2|$$

$$(ix) \quad |z_1 - z_2| \geq |z_1| - |z_2|$$

$$(x) \quad |z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2)$$

$$(xi) \quad |z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2)$$

$$(xii) \quad |z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2 \operatorname{Re}(z_1 \bar{z}_2)$$

Argand Diagram:

The representation of complex numbers as points in a plane forms an Argand diagram. The plane on which complex numbers are represented is known as the complex plane or Argand's plane or Gaussian plane. The x-axis is called the real axis and y-axis the imaginary axis. The complex number $z = x + iy$ is known as the affix of the point (x, y) which it represents.

Polar Form of Complex Number:

If $z = x + iy$ is a complex number. Then $z = r(\cos \theta + i \sin \theta)$ is called polar or trigonometric form.

Comparing real and imaginary parts, we get

$$x = r \cos \theta \quad (i)$$

$$\text{and } y = r \sin \theta \quad (ii)$$

squaring (i) and (ii) and adding, we get

$$r^2 = x^2 + y^2 \text{ or } r = \sqrt{x^2 + y^2} = |z|$$

Thus, r is known and is equal to the modulus of the complex number z .

Substituting the value of r in (i) and (ii), we get

$$\cos \theta = \frac{x}{\sqrt{x^2 + y^2}} \text{ and } \sin \theta = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\text{Dividing (ii) by (i), we get } \tan \theta = \frac{y}{x}. \quad (iii)$$

The form $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$ of the complex number z is called exponential form.

Any value of θ satisfying (iii) is known as **amplitude or argument** of z and written as $\theta = \arg(z)$ or $\theta = \operatorname{amp} z$.

The unique value of θ such that $-\pi < \theta \leq \pi$ for which $x = r \cos \theta$ and $y = r \sin \theta$ is known as the **principal value of the argument**.

The general value of the argument is $(2n\pi + \theta)$, where n is an integer and θ is the principle value. If $x > 0, y > 0$ (i. e., z is in first quadrant), then

$$\operatorname{Arg} z = \theta = \tan^{-1} \left(\frac{y}{x} \right).$$

If $x < 0, y > 0$ (i. e., z is the second quadrant), then

$$\text{Arg } z = \theta = \pi - \tan^{-1} \left(\frac{y}{|x|} \right).$$

If $x < 0, y < 0$ (i. e., z is in third quadrant), then

$$\text{Arg } z = \theta = -\pi + \tan^{-1} \left(\frac{y}{x} \right)$$

If $x > 0, y < 0$ (i. e., z is in fourth quadrant), then

$$\text{Arg } z = \theta = -\tan^{-1} \left(\frac{|y|}{x} \right).$$

Note: Argument of the complex number 0 is not defined.

Properties of Argument:

$$(i) \quad \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$(ii) \quad \arg \left(\frac{z_1}{z_2} \right) = \arg z_1 - \arg z_2$$

$$(iii) \quad \arg \left(\frac{z}{z} \right) = 2 \arg z$$

$$(iv) \quad \text{If } \arg \left(\frac{z_2}{z_1} \right) = \theta, \text{ then } \arg \left(\frac{z_1}{z_2} \right) = 2k\pi - \theta \text{ where } k \in \mathbb{I}.$$

$$(v) \quad \arg \bar{z} = -\arg z$$

Logarithm of Complex Number:

$$\text{Log}(\alpha + i\beta) = \frac{1}{2} \log(\alpha^2 + \beta^2) + i \tan^{-1} \frac{\beta}{\alpha} \quad (\alpha \neq 0)$$

$$\text{Log}(i\beta) = \log \beta + \frac{i\pi}{2}$$

Eular's Formula:

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta, \quad i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = e^{i\pi/2}$$

$$\log i = \log e^{i\pi/2} = \frac{i\pi}{2}, \quad \log(\log i) = \log \left(\frac{i\pi}{2} \right)$$

$$= \log i + \log \left(\frac{\pi}{2} \right) = \frac{i\pi}{2} + \log(\pi/2)$$

De Moivre's Theorem:

If n is any integer, then

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

Some Important Results:

(i) If n is any real number, then $\cos n\theta + i \sin n\theta$ is one of the values of $(\cos \theta + i \sin \theta)^n$

(ii) $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

(iii) $(\cos \theta - i \sin \theta)^n = \cos n\theta - i \sin n\theta$

(iv) $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

(v) $\frac{1}{\cos \theta + i \sin \theta} = (\cos \theta + i \sin \theta)^{-1} = \cos \theta - i \sin \theta$

(vi) $(\sin \theta + i \cos \theta)^n \neq \sin n\theta + i \cos n\theta$

In fact, $(\sin \theta + i \cos \theta)^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$

(vii) $(\cos \theta + i \sin \phi)^n \neq (\cos n\theta + i \sin n\phi)$

(viii) $(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \dots (\cos \theta_n + i \sin \theta_n)$
 $= \cos (\theta_1 + \theta_2 + \dots + \theta_n) + i \sin (\theta_1 + \theta_2 + \dots + \theta_n)$

Roots of A Complex Number:

If $z = r(\cos \theta + i \sin \theta)$ and n is a positive integer, then

$$z^{1/n} = r^{1/n} \left[\cos \left(\frac{2k\pi + \theta}{n} \right) + i \sin \left(\frac{2k\pi + \theta}{n} \right) \right]$$

Where $k = 0, 1, 2, 3, \dots, (n-1)$.

Cube roots of unity

Let $z = 1^{1/3}$ or $z^3 - 1 = 0$

$\Rightarrow (z-1)(z^2 + z + 1) = 0$.

i. e., $z = 1$ ($z^2 + z + 1 = 0$).

Put $\omega = \frac{-1 + i\sqrt{3}}{2}$, then $\omega^2 = \frac{-1 - i\sqrt{3}}{2}$. Thus cube roots of unity are $1, \omega, \omega^2$.

Properties of Cube Roots of Unity:

(i) $1 + \omega + \omega^2 = 0$

(ii) $\omega^3 = 1$

(iii) $\omega^{3n} = 1, \omega^{3n+1} = \omega, \omega^{3n+2} = \omega^2$

$\overline{\omega} = \omega^2$ and $(\overline{\omega})^2 = \omega$,

(iv) $\omega \overline{\omega} = \omega^3, \omega = e^{2\pi i/3}, \omega^2 = e^{4\pi i/3}$

(v) The cube roots of unity lie on the unit circle and divide the circumference into three equal parts

(vi) If $a + b\omega + c\omega^2 = 0$, then

$a = b = c$ provided a, b, c are real.

nth Roots of Unity:

Since $1 = \cos 0 + i \sin 0$, therefore

$$\begin{aligned}
 (1)^{1/n} &= (\cos 0 + i \sin 0)^{1/n} \\
 &= \cos \frac{2\pi r + 0}{n} + i \sin \frac{2\pi r + 0}{n}; r = 0, 1, 2, \dots, (n-1) \\
 &= \cos \frac{2\pi r}{n} + i \sin \frac{2\pi r}{n}; r = 0, 1, 2, \dots, (n-1) \\
 &= e^{i2\pi r/n}, e^{i4\pi/n}, \dots, e^{i2(n-1)\pi/n} \\
 &= 1, e^{i2\pi/n}, e^{i4\pi/n}, \dots, e^{i2(n-1)\pi/n}, \dots \\
 &= 1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1}, \text{ where } \alpha = e^{i2\pi/n}
 \end{aligned}$$

Properties of nth Roots of Unity:

- (i) $1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} = 0$
- (ii) $\alpha \cdot \alpha \cdot \alpha^2 \cdot \dots \cdot \alpha^{n-1} = (-1)^{n-1}$
- (iii) The n , n th roots of unity lie on the unit circle $|z| = 1$ and form the vertices of a regular polygon of n sides.
- (iv) n th roots of unity form a G. P. with common ratio $e^{i2\pi/n}$.

Equation Of Straight Line:

Equation of straight line through z_1 and z_2 is given by

$$\frac{z - z_1}{z_2 - z_1} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1}$$

Or
$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0$$

The general equation of straight line is

$$\bar{a}z + a\bar{z} + b = 0, \text{ where } b \text{ is a real number.}$$

Equation of the Perpendicular Bisector:

The equation of a circle with centre z_0 and radius r is $|z - z_0| = r$.

The general equation of a circle is $z\bar{z} + a\bar{z} + \bar{a}z + b = 0$, where b is a real number. The centre of this circle is $-a$ and its radius is $\sqrt{a\bar{a} - b}$.

Some Important Results:

(i) $\left| \frac{z - z_1}{z - z_2} \right| = k$ is a circle if $k \neq 1$ and is a line if $k = 1$.

(ii) The equation $|z - z_1|^2 + |z - z_2|^2 = k$ represents a circle if $k \geq \frac{1}{2} |z_1 - z_2|^2$.

(iii) If $\arg \left[\frac{(z_2 - z_3)(z_1 - z_4)}{(z_1 - z_3)(z_2 - z_4)} \right] = \pm \pi, 0$, then the points z_1, z_2, z_3, z_4 , are concyclic.

(iv) $|z - z_0| < r$ represents interior of the circle $|z - z_0| = r$ and

$|z - z_0| > r$ represents exterior of the circle $|z - z_0| = r$.

Equation of Ellipse:

If $|z - z_1| + |z - z_2| = 2a$, where $2a > |z - z_2|$, then the point z describe on the ellipse having foci at z_1 and z_2 and $a \in \mathbb{R}^+$.

Multiple Choice Questions:

- Which of the following is a set of complex number.
 - $\{a + ib \mid a, b \in \mathbb{R}\}$
 - $\{a + ib \mid a, b \in \mathbb{C} \wedge i = \sqrt{-1}\}$
 - $\{a + ib \mid a, b \in \mathbb{Z} \wedge i^2 = -1\}$
 - $\{a + ib \mid a, b \in \mathbb{R} \wedge i^2 + 1 = 0\}$
- The set of complex number is a/an
 - Abelian group w.r to addition
 - Ring
 - Field
 - All of these
- If $Z = 3 - 4i$ then conjugate of Z lies in
 - Quad-I
 - Quad-II
 - Quad-III
 - Quad-IV
- Which of the following is polar form of $1 - i\sqrt{3}$.
 - $2 (\cos 60^\circ - i \sin 60^\circ)$
 - $2 (\cos 90^\circ - i \sin 90^\circ)$
 - $3 (\cos 45^\circ + i \sin 45^\circ)$
 - $4 (\cos 30^\circ + i \sin 30^\circ)$
- If $Z = 2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$ then $Z =$
 - $2 - i\sqrt{3}$
 - $-5i$
 - $-1 + 2i$
 - $-2 + 2i$
- $\arg \left(\frac{Z_1}{Z_2} \right) =$
 - $\arg (Z_1 - Z_2)$
 - $\arg (Z_1) + \arg (Z_2)$
 - $\arg (Z_1) - \arg (Z_2)$
 - $\arg (Z_1) \arg (Z_2)$
- The greatest value of $|Z_1 + Z_2|$ is:
 - $|Z_1 - Z_2|$
 - $|Z_1 Z_2|$
 - $||Z_1| - |Z_2||$
 - $|Z_1| + |Z_2|$
- Which of the following statement is true if Z is a complex number.
 - $Z + \bar{Z} = 2 \operatorname{Re} (Z)$
 - $Z - \bar{Z} = 2 \operatorname{Im} (Z)$
 - $Z \bar{Z} = |Z|^2$
 - All of these
- Which of the following statement is not true if Z is a complex number.
 - $(\operatorname{Re} (Z))^2 \leq |Z|^2$
 - $(\operatorname{Im} (Z))^2 \leq |Z|^2$
 - $Z^{-1} |Z|^2 = \bar{Z}$
 - None
- If $W = f(Z)$ such that for given value of Z there correspond more than one values of w . Then w is called
 - Relation
 - Single valued function
 - Multivalued function
 - Not a function
- Which of the following is a multivalued function.
 - $W = Z^2$
 - $W = \sqrt{Z}$
 - $W = \sin (Z)$
 - $W = \log (Z)$

12. Which of the following is a single valued function.
 (a) $W = Z^3$ (b) $W = Z^{1/3}$
 (c) $W = Z^{1/n}$ (d) $W = \pm Z^2$
13. A function $W = f(Z)$ has limit W_0 as Z tends to Z_0 if for given ϵ there is $\delta > 0$ such that $|f(Z) - W_0| < \epsilon$ whenever
 (a) $Z - Z_0 < \delta$ (b) $|Z - Z_0| < \delta$
 (c) $0 < |Z - Z_0| < \delta$ (d) $Z - \delta < Z_0$
14. $\lim_{Z \rightarrow 1} \left(\frac{Z^2 - 1}{Z - 1} \right) =$
 (a) 1 (b) 2 (c) 3 (d) 4
15. $\lim_{Z \rightarrow 1} \left(\frac{Z^n - 1}{Z - 1} \right) =$
 (a) 1 (b) 2 (c) n (d) 4
16. $\lim_{Z \rightarrow 2i} \frac{Z^2 + 4}{Z - 2i} =$
 (a) 1 (b) $2i$ (c) $4i$ (d) 4
17. Let $W = f(Z)$ be a complex valued function such that $|f(Z) - f(Z_0)| < \epsilon$ whenever $0 \leq |Z - Z_0| < \delta$ then $W = f(Z)$ is
 (a) multivalued function (b) continuous at Z_0
 (c) differentiable at Z_0 (d) none
18. The function $f(Z) = |Z|^2$ is continuous at
 (a) 0 (b) 1 (c) 10 (d) All of these
19. The function $f(Z) = \frac{Z + 4i}{Z^2 + 9}$ is discontinuous at:
 (a) 9 (b) $-3i$ (c) $4i$ (d) 0
20. The function $W = \frac{Z^2 + 3}{(Z^2 + 1)(Z^2 + 16)}$ is continuous at
 (a) i (b) $4i$ (c) $-4i$ (d) $3i$
21. Which of the following function is discontinuous at 5.
 (a) $W = \frac{Z + 2}{Z - 5}$ (b) $W = Z + 5i$
 (c) $W = \frac{Z}{Z + 5}$ (d) None
22. Which of the following statement is true.
 (a) A function is differentiable then it is also continuous
 (b) A function is uniformly continuous then it is also continuous
 (c) A function is uniformly continuous then it is also differentiation
 (d) None

23. At which point the function $W = \bar{Z}$ is continuous as well as differentiable.
(a) 0 (b) 1 (c) -1 (d) i
24. At which point the function $W = |Z|^2$ is continuous as well as differentiable.
(a) 0 (b) 1 (c) i (d) None
25. If Z is a complex number, then which of the following is differentiable at any point Z_0 .
(a) $\operatorname{Re}(Z)$ (b) $\operatorname{Im}(Z)$
(c) $\operatorname{Arg}(Z)$ (d) None
26. If $f(Z)$ is a single valued function which is differentiable in its domain, then $f(Z)$ is called
(a) analytic function (b) regular function
(c) holomorphic function (d) All of these
27. If $f(Z) = u(x, y) + iv(x, y)$ is analytic function then
(a) u_x, u_y, v_x and v_y are continuous (b) $f(Z)$ satisfies Cauchy Riemann equations
(c) $f(Z)$ satisfies $u_x = v_y$ and $u_y = -v_x$ (d) All of these
28. The function $W = f(Z)$ satisfies Cauchy-Riemann equations. Then $W =$
(a) $|Z|^2$ (b) $f(\bar{Z})$
(c) k , k is a constant (d) None
29. If Cauchy Riemann equation holds for a function $f(Z)$, then $f(Z)$ is:
(a) analytic (b) not analytic
(c) may or may not be analytic (d) holomorphic
30. Which of the following function is analytic.
(a) $f(Z) = 2Z$ (b) $f(Z) = Z + \bar{Z}$
(c) $f(Z) = \sin(16x + 24iy)$ (d) $f(Z) = Z\bar{Z}$
31. A function is analytic if it is a function of
(a) Z alone (b) $\bar{Z}$ alone
(c) Z^2 alone (d) $|Z|$ alone
32. A real valued function $U(x, y)$ satisfies Laplace equation i.e., $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = \nabla^2 U = 0$. The $U(x, y)$ is called:
(a) Harmonic function (b) Regular function
(c) Analytic function (d) None
33. Which of the following is an analytic function.
(a) $f(Z) = \sin(37x + 35iy)$ (b) $f(Z) = \cos(7x + 5iy)$
(c) $f(Z) = \sin(x - iy)$ (d) none
34. Which of the following is correct.
(a) $\sin(iz) = i \sin h(z)$ (b) $\cos(iz) = \cos h(z)$
(c) $\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$
(d) all of these

35. If $Z = x + iy$ then $|\sin z|^2 =$
 (a) $\sin x + \sin y$ (b) $\sin^2 x + \sin^2 y$
 (c) $\sin^2 x + \sin^2 y$ (d) $\sin^2 x + \cos^2 y$
36. If $z = x + iy$, then $|\cos z|^2 =$
 (a) $\cos^2 x + \cos^2 y$ (b) $\cos^2 x + \cosh^2 y$
 (c) $\cos^2 x + \cosh^2 y$ (d) $\cos^2 x + \sinh^2 y$
37. If $Z = x + iy$, then $|\cos hz|^2 =$
 (a) $\cos^2 y + \sinh^2 y$ (b) $\cos^2 y + \sinh^2 x$
 (c) $\sinh^2 x + \cosh^2 y$ (d) $\sin^2 x + \cos^2 x$
38. If Z is a complex number, then $\log Z =$
 (a) $\log |Z| + i \arg(z)$ (b) $\sqrt{x^2 + y^2}$
 (c) $\sin z + \cos z$ (d) $\tan^{-1}(z)$
39. $\log(i) =$
 (a) 1 (b) $i\frac{\pi}{2}$ (c) 0 (d) e^{iz}
40. $\log(i - 1) =$
 (a) $\log 2 + i\frac{\pi}{2}$ (b) $\frac{1}{2} \log 2 + i\frac{\pi}{3}$
 (c) $\frac{1}{2} \log 2 + i\frac{3\pi}{4}$ (d) $\log 2 + i\frac{\pi}{4}$
41. $\log(\sqrt{3} + i) =$
 (a) $\log 2 + i\frac{\pi}{6}$ (b) $\log 3 + i\frac{\pi}{2}$ (c) $i\frac{\pi}{4}$ (d) none
42. $\log(1 - i) =$
 (a) $\log \sqrt{2} + i\frac{\pi}{4}$ (b) $\frac{1}{2} \log 2 + i\frac{\pi}{4}$
 (c) $\frac{1}{2} \log 2 - i\frac{\pi}{4}$ (d) $\log 2 + i\frac{\pi}{4}$
43. $e^{\log(1+i)} =$
 (a) 1 (b) $1 + i$ (c) $1 - i$ (d) e^i
44. $e^{iz} =$
 (a) $\cos z + i \sin z$ (b) $\cos z - i \sin z$
 (c) $\cos z - \sin z$ (d) $\cos hz + \sin hz$
45. $(-i)^i =$
 (a) 1 (b) 0 (c) $e^{\pi/2}$ (d) $-i$
46. $(-1)^{2i} =$
 (a) 1 (b) $e^{-2\pi}$ (c) $e^{\pi/2}$ (d) 0
47. $e^{-\pi/4} + i\frac{\log 2}{2} =$

- (a) i (b) $1 + i$ (c) $(1 + i)^i$ (d) $(i)^i$
48. $(i)^i =$
 (a) $e^{-\pi/2}$ (b) $e^{\pi/2}$ (c) 0 (d) 1
49. A curve traced by complex valued function $w = f(z)$ such that $f(z_1) \neq f(z_2)$ for $z_1 \neq z_2$ then curve is called:
 (a) smooth curve (b) simple curve
 (c) Jordan curve (d) closed curve
50. A curve which is simple as well as closed curve is called:
 (a) smooth curve (b) simple curve
 (c) Jordan curve (d) closed curve
51. If $\int_C Z^n dz = 0$, then C is
 (a) simple curve (b) smooth curve
 (c) closed curve (d) analytic curve
52. If C is a circle with centre at Z_0 and radius r , then $\int_C (Z - Z_0) dZ =$
 (a) 1 (b) $2\pi i$ (c) 0 (d) πi
53. If C is a circle with centre at Z_0 and radius r , then $\int_C (Z - Z_0)^{-1} dZ =$
 (a) 1 (b) $2\pi i$ (c) 0 (d) πi
54. Piece wise smooth curve is also called:
 (a) regular curve (b) contour
 (c) smooth curve (d) circle
55. $\left| \int_C f(Z) dZ \right| \leq |f(Z)| L$ where $L =$
 (a) limit of C (b) length of contour C
 (c) length of circle C (d) none
56. $\left| \int_i^{2+i} \frac{1}{Z^2} dZ \right| \leq$
 (a) 1 (b) 2 (c) 2π (d) $-\pi i$
57. If C is a semi unit circle, then $\left| \int_C \bar{Z} dZ \right| \leq$
 (a) 1 (b) 2 (c) π (d) 2π
58. $\int_{\pi}^0 \frac{1}{Z} dZ =$
 (a) 1 (b) 0 (c) πi (d) $-\pi i$

59. If C is a straight line from $Z = -i$ to $Z = i$ then $\int_C |Z| dZ =$
 (a) 0 (b) 1 (c) i (d) π
60. If C is a circle $|Z| = 2$ then $\int_C \frac{dZ}{Z^2 - 1} =$
 (a) 0 (b) 1 (c) i (d) 2π
61. If C is a semi-circle $Z = \pm i$ as ends of the diameter, then $\left| \int_C (x^2 + iy^2) dZ \right| \leq$
 (a) i (b) π (c) 2π (d) $\frac{\pi}{2}$
62. Let $f(Z)$ be analytic in a simply connected domain D and let C be any closed continuous curve in D then $\int_C f(Z) dZ = 0$ is a statement of:
 (a) Cauchy fundamental theorem (b) Cauchy Goursat theorem
 (c) Cauchy Riemann theorem (d) none
63. If C is a unit circle such that $\int_C f(Z) = 0$ then $f(Z) =$
 (a) $f(Z) = \frac{1}{Z^2}$ (b) $f(Z) = \frac{\sin hZ}{Z^2 - 4}$
 (c) $f(Z) = \frac{\cos hZ}{Z + 5}$ (d) all of these
64. If C is a closed contour enclosing the point $Z = a$ then $\int_C \frac{dZ}{(Z - a)^5} =$
 (a) 0 (b) 1 (c) $2\pi i$ (d) i
65. If C is a closed contour enclosing the point $Z = a$ then $\int_C \frac{dZ}{(Z - a)} =$
 (a) 0 (b) 1 (c) $2\pi i$ (d) i
66. If C is a closed contour enclosing point $Z = a$ then $\int_C \frac{dZ}{(Z - a)^2} =$
 (a) 0 (b) 1 (c) $2\pi i$ (d) i
67. If $C : |Z| = 1$ then $\int_C \frac{\cos 2Z + \cos h2Z}{Z} =$
 (a) πi (b) $2\pi i$ (c) $3\pi i$ (d) $4\pi i$

68. Let $f(Z)$ be analytic on C of a simply connected domain D and let $a \in C$ then $\int_C \frac{f(Z)}{(Z-a)^2} dZ = kf'(a)$ where $k =$
- (a) πi (b) $2\pi i$ (c) $3\pi i$ (d) $4\pi i$
69. Let $f(Z)$ be analytic on C of a simply connected domain D such that $a \in C$ then $\int_C \frac{f(Z)}{(Z-a)^4} dZ = \frac{2\pi i}{3!} \times \text{---}$
- (a) $f'(a)$ (b) $f''(a)$
 (c) $f'''(a)$ (d) $f^{IV}(a)$
70. If $f(Z)$ be analytic on a closed contour $C : |Z - a| = r$ such that $|f(Z)| \leq M$ then which of the following is Cauchy's Inequality.
- (a) $|f(a)| \leq n! M$ (b) $|f(a)| \leq \frac{n!}{r^n} M$
 (c) $|f^{(n)}(a)| \leq \frac{n!}{r^n} M$ (d) $|f'(a)| \leq n! M$
71. A function which is analytic every where in the complex plane is called:
- (a) harmonic function (b) regular function
 (c) entire function (d) differentiable function
72. Which of the following is entire function.
- (a) $\sin Z$ (b) e^Z
 (c) $\cos hZ$ (d) all of these
73. Which of the following is entire function.
- (a) polynomial function (b) algebraic function
 (c) transcendental function (d) none
74. Every entire bounded function is
- (a) polynomial (b) constant
 (c) monomial (d) binomial
75. If $f(a+h) = a_0 + a_1h + \dots + a_nh^n + \frac{b_1}{h} + \dots + \frac{b_n}{h_n} + \dots$ is a Laurent series of $f(Z)$ then $a_n =$
- (a) $\frac{1}{2\pi i} \int_C f(Z) dZ$ where $n = 0, 1, 2, \dots$
 (b) $\frac{1}{2\pi i} \int_C \frac{f(Z)}{(Z-a)^{n+1}} dZ$ where $n = 0, 1, 2, \dots$
 (c) $\frac{1}{2\pi i} \int_C f(Z) (Z-a)^{n-1} dZ$ where $n = 1, 2, \dots$
 (d) none
76. If $f(Z) = \frac{1}{(Z+1)(Z+3)}$ then $f(Z) = \frac{1}{3} - \frac{4}{9}Z + \frac{13}{27}Z^2 - \frac{40}{81}Z^3 + \dots$ is a

- (a) Taylor's series (b) Laurent's series
(c) Power series (d) none
77. Let $f(Z)$ be a complex valued function defined on D such that $a \in D$, if $f(a) = 0$ then $Z = a$ is called:
(a) zero of $f(Z)$ (b) singularity of $f(Z)$
(c) residue of $f(Z)$ (d) none
78. If a function $f(Z)$ is not analytic at $Z = a$ then $Z = a$ is called:
(a) zero of $f(Z)$ (b) singularity of $f(Z)$
(c) isolated singularity (d) pole
79. Which of the following is a kind of singularity.
(a) pole (b) removable singularity
(c) isolated singularity (d) all of these
80. A function which has poles as its only singularities in the finite part of the plane is called:
(a) monogenic function (b) meromorphic function
(c) analytic function (d) none
81. Which of the following function has isolated essential singularity at $Z = 0$.
(a) $f(Z) = \frac{e^Z}{(Z-1)^3}$ (b) $f(Z) = \frac{Z - \sin Z}{Z^3}$
(c) $f(Z) = e^{1/2Z}$ (d) none
82. A part of a many valued complex function which is single valued and analytic is called:
(a) point (b) branch
(c) cut (d) section

KEY To MCQ Of Complex Analysis

1	d	2	d	3	a	4	a	5	d	6	c
7	d	8	d	9	d	10	c	11	d	12	a
13	c	14	a	15	c	16	?	17	b	18	d
19	b	20	d	21	a	22	a	23	a	24	d
25	d	26	d	27	d	28	d	29	c	30	a
31	a	32	a	33	d	34	d	35	c	36	d
37	b	38	a	39	b	40	c	41	a	42	c
43	b	44	a	45	c	46	b	47	c	48	a
49	b	50	c	51	c	52	c	53	b	54	b
55	b	56	b	57	c	58	c	59	c	60	a
61	b	62	b	63	d	64	a	65	c	66	a
67	d	68	b	69	c	70	c	71	c	72	d
73	c	74	b	75	b	76	b	77	a	78	b
79	d	80	b	81	c	82	b				